

SDN INTEROPERABILITY IN HYBRID ENVIRONMENTS

Bridging Legacy Non-OpenFlow Devices and OpenFlow Switches

OVERVIEW

During migration from a traditional network to a Software-Defined Network (SDN), SDN controllers manage interoperability between legacy devices and OpenFlow-enabled switches through hybrid control architectures, multi-protocol southbound interfaces, standard protocol translation, and overlay tunneling.

PRIMARY STRATEGIES FOR HYBRID INTEROPERABILITY

- **Multi-Protocol Southbound Plugins:** Modern SDN controllers (such as OpenDaylight or ONOS) use pluggable southbound drivers. The controller communicates with OpenFlow switches using the OpenFlow protocol while concurrently interfacing with legacy network devices via traditional management and control protocols like SNMP, NETCONF, CLI/SSH, BGP, or OSPF.
- **Hybrid Switch Modes (Hybrid Ports / Ships-in-the-Night):** Hybrid OpenFlow switches support dual-pipeline processing. Packets matching explicit OpenFlow entries are processed by the SDN controller, whereas non-matching or traditional traffic bypasses OpenFlow processing and is passed to the legacy hardware switching ASIC running standard L2/L3 protocols (OSPF, STP, LACP).
- **Protocol Translation and Peering:** The SDN controller acts as a bridge between the OpenFlow domain and the traditional control plane. For example, the controller can run BGP or OSPF speakers internally to peer with legacy edge routers, translate standard route advertisements into OpenFlow flow modifications, and inject them into OpenFlow switches.
- **Overlay Tunneling & Encapsulation:** Controllers instantiate overlay tunnels (e.g., VXLAN, GRE) between SDN-managed virtual/physical switches across an underlay network composed of legacy hardware switches. To legacy devices, SDN-encapsulated traffic appears as standard IP unicast packets, allowing seamless transport without requiring OpenFlow support on intermediate hardware.
- **Unified Topology Mapping:** The controller builds a single consolidated global network view by synthesizing topology information gathered from OpenFlow LLDP messages with legacy discovery mechanisms like SNMP, CDP, and LLDP from non-OpenFlow nodes.

CONCLUSION

By integrating legacy protocol drivers, peering with traditional routing protocols, using hybrid switch pipelines, and leveraging overlay encapsulation, SDN controllers ensure seamless end-to-end connectivity during network transitions.